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Dept. Name : CSE -AIDS

Course Code/Course Name: DSA06 /Data Handling and Visualization

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LIST OF EXPERIMENTS

Set 7 Customer Demographics Analysis

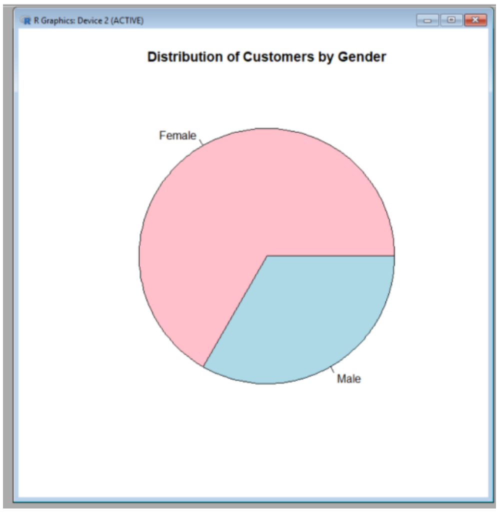
Dataset: Customer Demographics

1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

```
# Customer Age Data
age <- c(28, 35, 42)

# Customer IDs
customer <- c("Customer 1", "Customer 2", "Customer 3")

# Create Bar Chart
barplot(age,
    names.arg = customer,
    main = "Distribution of Customer Ages",
    xlab = "Customers",
    ylab = "Age (Years)",
    col = "skyblue",
    border = "black")
```

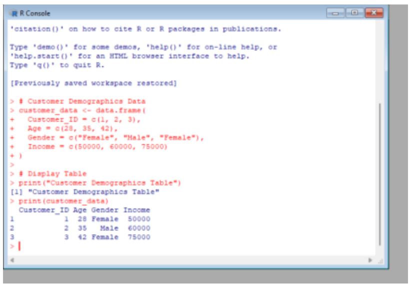
2. Generate a pie chart to represent the distribution of customers by gender.

```
# Customer Gender Data

gender <- c("Female", "Male", "Female")


# Count the number of customers by gender

gender_count <- table(gender)


# Create Pie Chart

pie(gender_count,

   labels = names(gender_count),

   main = "Distribution of Customers by Gender",

   col = c("pink", "lightblue"))
```



3. Build a table to show the demographic information for each customer. Label the table elements.

```
# Customer Demographics Data
customer_data <- data.frame(
  Customer_ID = c(1, 2, 3),
  Age = c(28, 35, 42),
  Gender = c("Female", "Male", "Female"),
  Income = c(50000, 60000, 75000)
)

# Display Table
print("Customer Demographics Table")
print(customer_data)
```



4. Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

Steps to Create the Dashboard

1. Open **Tableau Desktop**.
2. Import the **Customer Demographics** dataset (Excel/CSV).

Sheet 1: Bar Chart

- Drag **Customer ID** (or Customer Name) to **Columns**.
- Drag **Age** to **Rows**.
- Select **Bar** from the **Marks** card.
- Add the title **"Customer Age Distribution"**.

Sheet 2: Pie Chart

- Drag **Gender** to **Color**.
- Drag **Gender** to **Label**.
- Drag **Number of Records** to **Angle**.
- Select **Pie** from the **Marks** card.
- Add the title **"Customer Distribution by Gender"**.

Sheet 3: Customer Demographics Table

- Create a new worksheet.
- Drag **Customer ID** to **Rows**.
- Drag **Age**, **Gender**, and **Income** to **Text**.
- Add the title **"Customer Demographics Table"**.

Create the Dashboard

1. Click **Dashboard → New Dashboard**.
2. Drag the **Bar Chart**, **Pie Chart**, and **Customer Demographics Table** onto the dashboard.
3. Arrange them neatly for easy viewing.

Make the Dashboard Interactive

- Click **Dashboard → Actions → Add Action → Filter**.
- Set any chart as the **Source** and the remaining charts/table as the **Target**.
- Choose **Run action on Select**.
- Clicking on a bar or pie slice will filter and update the other visualizations.

Dashboard Features

- 📊 **Bar Chart** showing customer age distribution.
- 🥧 **Pie Chart** showing gender distribution.
- 📋 **Table** displaying customer demographic details.
- 🔍 Interactive filtering and highlighting for easy exploration of customer demographics.

Set 8 Employee Performance Analysis

Dataset: Employee Performance

1. Create a line chart to visualize the performance trend of employees over time. Label the axes and

title the chart.

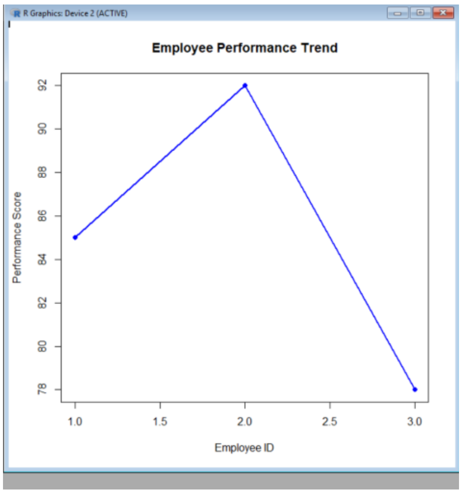
Employee Data

employee <- c(1, 2, 3)

performance <- c(85, 92, 78)

Create Line Chart

```
plot(employee, performance,
    type = "o",
    col = "blue",
    pch = 16,
    lwd = 2,
    main = "Employee Performance Trend",
    xlab = "Employee ID",
    ylab = "Performance Score")

# Add Legend
legend("topright",
      legend = "Performance Score",
      col = "blue",
      lty = 1,
      pch = 16)
```



2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

```
# Employee Department Data
department <- c("Sales", "HR", "Marketing")

# Count Employees in Each Department
dept_count <- table(department)

# Create Bar Chart
barplot(dept_count,
        main = "Employee Distribution by Department",
        xlab = "Department",
        ylab = "Number of Employees",
        col = c("skyblue", "lightgreen", "orange"),
        border = "black")
```

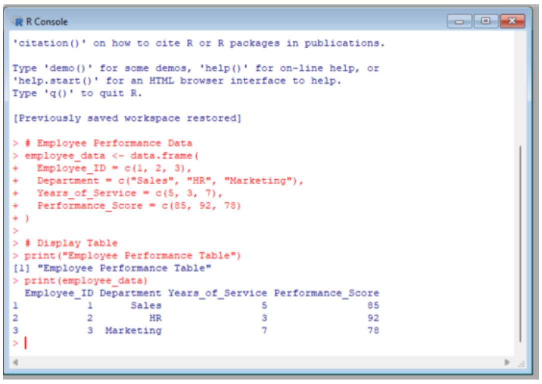
3.Build a table to display the performance data for each employee. Label the table elements.

```
# Employee Performance Data
employee_data <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)
```

Display Table

print("Employee Performance Table")

print(employee_data)



4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.

Steps to Create the Dashboard

1. Open **Tableau Desktop**.
2. Import the **Employee Performance** dataset (Excel/CSV).

Sheet 1: Line Chart

- Drag **Employee ID** to **Columns**.
- Drag **Performance Score** to **Rows**.
- Select **Line** from the **Marks** card.
- Add the title **"Employee Performance Trend"**.

Sheet 2: Bar Chart

- Drag **Department** to **Columns**.
- Drag **Number of Records** (or COUNT(Employee ID)) to **Rows**.
- Select **Bar** from the **Marks** card.

- Add the title **"Employee Distribution by Department"**.

Sheet 3: Employee Performance Table

- Create a new worksheet.
- Drag **Employee ID** to **Rows**.
- Drag **Department**, **Years of Service**, and **Performance Score** to **Text**.
- Add the title **"Employee Performance Table"**.

Create the Dashboard

1. Click **Dashboard → New Dashboard**.
2. Drag the **Line Chart**, **Bar Chart**, and **Employee Performance Table** onto the dashboard.
3. Arrange them neatly for easy viewing.

Make the Dashboard Interactive

- Click **Dashboard → Actions → Add Action → Filter**.
- Set one visualization as the **Source** and the remaining visualizations as the **Target**.
- Choose **Run action on Select**.
- Clicking on an employee or department in one visualization will automatically filter and update the others.

Dashboard Features

- 📈 **Line Chart** showing employee performance trend.
- 📊 **Bar Chart** showing employee distribution by department.
- 📋 **Table** displaying employee performance details.
- 🔍 Interactive filtering, highlighting, and comparison of employee performance data.

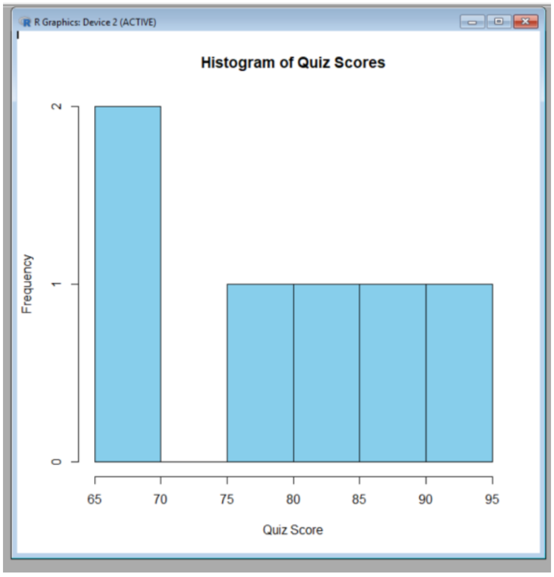
set 9

Dataset: Online_Learning_Activity.csv

1. Import the dataset in R. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by Course. Add title and labels. Interpret student performance.

Create the dataset

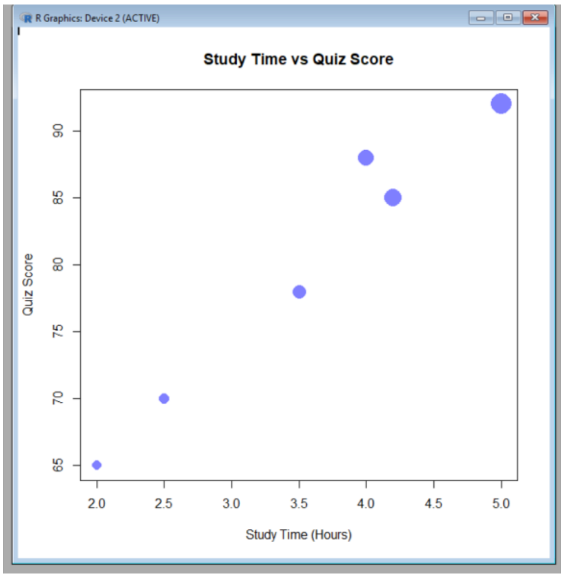
```r
student_data <- data.frame(
  Student_ID = c("L01","L02","L03","L04","L05","L06"),
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Age = c(20,22,19,21,23,20),
  Course = c("R","R","SQL","R","R","SQL"),
  Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),
  Videos_Watched = c(12,15,8,18,9,14),
  Quiz_Score = c(78,85,65,92,70,88),
  Login_Date = as.Date(c("2025-01-05","2025-01-05",
                         "2025-02-08","2025-02-08",
                         "2025-03-12","2025-03-12"))
)

# Histogram
hist(student_data$Quiz_Score,
     main = "Histogram of Quiz Scores",
     xlab = "Quiz Score",
     ylab = "Frequency",
     col = "skyblue",
     border = "black")

# Boxplot
boxplot(Quiz_Score ~ Course,
        data = student_data,
        main = "Quiz Score by Course",
        xlab = "Course",
        ylab = "Quiz Score",
        col = c("lightgreen", "orange"))
```



2. Create a scatter plot using R programming between Study_Time and Quiz_Score. Represent Videos_Watched using bubble size. Apply transparency. Explain the learning pattern.

```
# Create the dataset
student_data <- data.frame(
  Student_ID = c("L01","L02","L03","L04","L05","L06"),
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Age = c(20,22,19,21,23,20),
  Course = c("R","R","SQL","R","R","SQL"),
  Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),
  Videos_Watched = c(12,15,8,18,9,14),
  Quiz_Score = c(78,85,65,92,70,88)
)

# Bubble Scatter Plot
plot(student_data$Study_Time,
     student_data$Quiz_Score,
     cex = student_data$Videos_Watched/5,
```

```
  pch = 16,
  col = rgb(0, 0, 1, 0.5),
  main = "Study Time vs Quiz Score",
  xlab = "Study Time (Hours)",
  ylab = "Quiz Score")
```



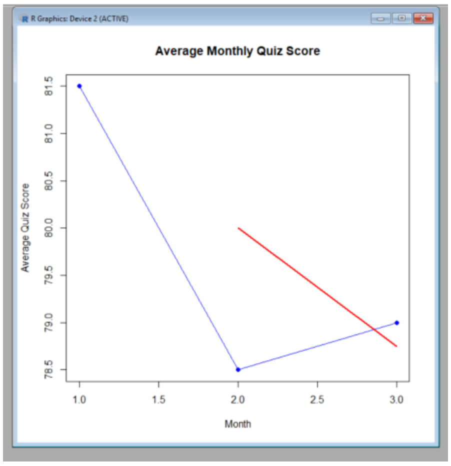
3. Using R programming, convert Login_Date into Date format. Calculate average quiz score per month. Plot a line chart. Apply moving average smoothing. Identify the trend.

```
# Create the dataset
student_data <- data.frame(
  Student_ID = c("L01","L02","L03","L04","L05","L06"),
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Age = c(20,22,19,21,23,20),
  Course = c("R","R","SQL","R","R","SQL"),
  Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),
  Videos_Watched = c(12,15,8,18,9,14),
  Quiz_Score = c(78,85,65,92,70,88),
```

```
  Login_Date = c("2025-01-05","2025-01-05",
                 "2025-02-08","2025-02-08",
                 "2025-03-12","2025-03-12")
)

# Convert Login_Date to Date format
student_data$Login_Date <- as.Date(student_data$Login_Date)

# Extract Month
month <- format(student_data$Login_Date, "%Y-%m")

# Calculate Average Quiz Score per Month
avg_score <- tapply(student_data$Quiz_Score, month, mean)

# Plot Line Chart
plot(avg_score,
     type = "o",
     col = "blue",
     pch = 16,
     xlab = "Month",
     ylab = "Average Quiz Score",
     main = "Average Monthly Quiz Score")

# Moving Average (2-point)
ma <- filter(avg_score, rep(1/2, 2), sides = 1)

# Add Moving Average Line
lines(ma, col = "red", lwd = 2)
```

Legend

```
legend("topleft",
       legend = c("Average Score", "Moving Average"),
       col = c("blue", "red"),
       lty = 1,
       pch = 16)
```



4. Import Online_Learning_Activity.csv into Tableau. Create a bar chart showing average quiz score by Course. Create a scatter plot (Study Time vs Quiz Score). Add filter for Gender and Course. Design a mini interactive dashboard.

Steps to Create the Dashboard

1. Open **Tableau Desktop**.
2. Import the **Online_Learning_Activity.csv** dataset (or create the dataset in Excel and save it as CSV if your question paper only provides the data).

Sheet 1: Bar Chart (Average Quiz Score by Course)

- Drag **Course** to **Columns**.
- Drag **Quiz_Score** to **Rows**.
- Click the drop-down on **SUM(Quiz_Score)** and change it to **Average (AVG)**.
- Select **Bar** from the **Marks** card.

- Add the title **"Average Quiz Score by Course"**.

Sheet 2: Scatter Plot (Study Time vs Quiz Score)

- Drag **Study_Time (hrs)** to **Columns**.
- Drag **Quiz_Score** to **Rows**.
- Select **Circle** from the **Marks** card.
- (Optional) Drag **Videos_Watched** to **Size** to vary the bubble size.
- Add the title **"Study Time vs Quiz Score"**.

Add Filters

- Drag **Gender** to the **Filters** shelf and choose **Show Filter**.
- Drag **Course** to the **Filters** shelf and choose **Show Filter**.

Create the Dashboard

1. Click **Dashboard → New Dashboard**.
2. Drag the **Bar Chart** and **Scatter Plot** onto the dashboard.
3. Place the **Gender** and **Course** filter cards on the side of the dashboard.

Make the Dashboard Interactive

- Click **Dashboard → Actions → Add Action → Filter**.
- Select the charts as both **Source** and **Target**.
- Choose **Run action on Select**.
- Clicking a course in the bar chart filters the scatter plot, and the **Gender** and **Course** filters update both charts.

Dashboard Features

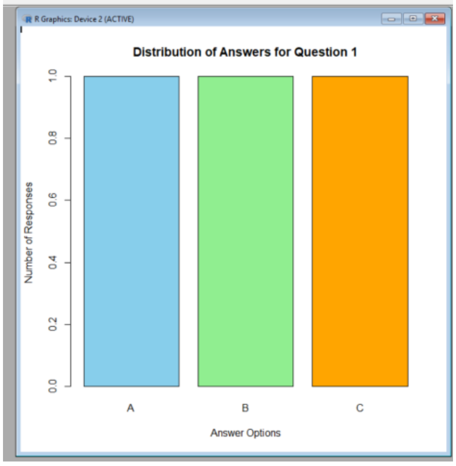
- 📊 **Bar Chart** showing the **Average Quiz Score by Course**.
- 🔵 **Scatter Plot** showing **Study Time vs Quiz Score**.
- 🎛️ Interactive **Gender** and **Course** filters.
- 🔍 Users can interactively explore student learning patterns and compare performance across different courses and genders.

Set 10 Survey Responses Analysis

Dataset: Survey Response0073

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

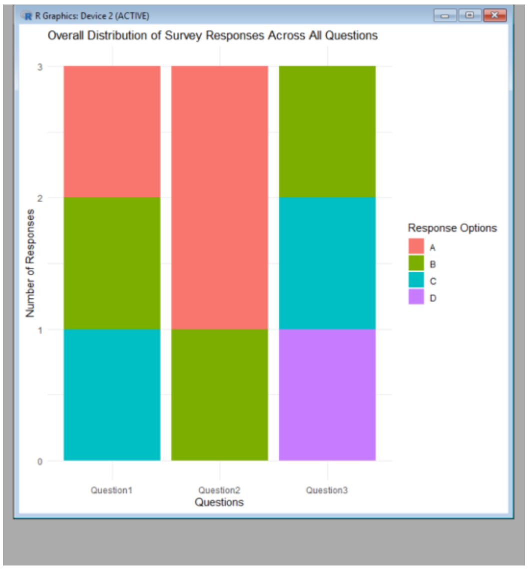
```
# Create the dataset
survey_data <- data.frame(
  Survey_ID = c(1, 2, 3),
  Question1 = c("A", "B", "C"),
  Question2 = c("B", "A", "A"),
  Question3 = c("C", "D", "B")
)

# Count responses for Question 1
q1_count <- table(survey_data$Question1)

# Create Grouped Bar Chart
barplot(q1_count,
        beside = TRUE,
        main = "Distribution of Answers for Question 1",
        xlab = "Answer Options",
        ylab = "Number of Responses",
        col = c("skyblue", "lightgreen", "orange"),
        border = "black")
```



2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

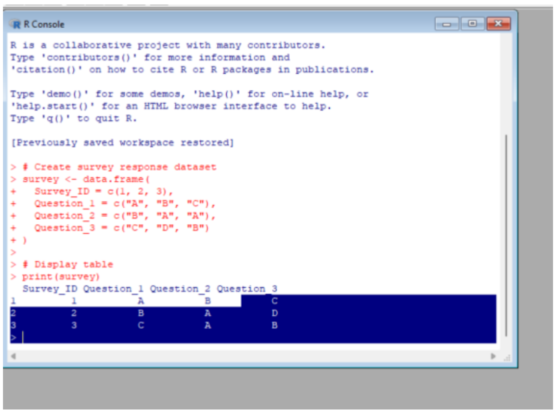
```
# Create dataset
survey <- data.frame(
  Survey_ID = c(1, 2, 3),
  Question1 = c("A", "B", "C"),
  Question2 = c("B", "A", "A"),
  Question3 = c("C", "D", "B")
)

# Load library
library(ggplot2)
library(tidyr)
library(dplyr)

# Convert data from wide format to long format
survey_long <- survey %>%
  pivot_longer(
```

```
    cols = Question1:Question3,
    names_to = "Question",
    values_to = "Response"
  )

# Count responses
response_count <- survey_long %>%
  group_by(Question, Response) %>%
  summarise(Count = n())

# Create stacked bar chart
ggplot(response_count, aes(x = Question, y = Count, fill = Response)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Overall Distribution of Survey Responses Across All Questions",
    x = "Questions",
    y = "Number of Responses",
    fill = "Response Options"
  ) +
  theme_minimal()
```



4. **Build a table to show the survey response data for each survey. Label the table elements.**

```
# Create survey response dataset
survey <- data.frame(
  Survey_ID = c(1, 2, 3),
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "A", "A"),
  Question_3 = c("C", "D", "B")
)

# Display table
print(survey)
```



4. Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.

Steps to Create Dashboard in Tableau:

1. **Import Dataset**
   - Open Tableau Desktop.
   - Connect to the **Survey Responses** dataset.
2. **Create Worksheets**

Worksheet 1: Grouped Bar Chart

- Drag **Question 1** to Columns.
- Drag **Survey ID** to Rows and select **Count**.
- Select **Bar Chart**.
- Add **Question 1** to Color.
- Title: **"Distribution of Answers for Question 1"**

Worksheet 2: Stacked Bar Chart

- Convert Question columns into rows using **Pivot**.
- Drag **Question** to Columns.
- Drag **Response** to Color.
- Drag **Response** to Rows and select **Count**.
- Select **Stacked Bar Chart**.

- Title: **"Overall Distribution of Responses Across Questions"**

Worksheet 3: Survey Response Table

- Drag **Survey ID**, **Question 1**, **Question 2**, and **Question 3** into Rows.
- Select **Text Table**.
- Title: **"Survey Response Details"**

Creating the Dashboard:

1. Click **New Dashboard**.
2. Set dashboard size (e.g., Automatic).
3. Drag the three worksheets into the dashboard:
    - Grouped Bar Chart
    - Stacked Bar Chart
    - Survey Response Table
4. Arrange the layout:
    - Top: Grouped Bar Chart
    - Middle: Stacked Bar Chart
    - Bottom: Survey Response Table

Add Interactivity:

Filters:

- Add **Survey ID** as a filter to allow users to view individual survey responses.

Actions:

- Go to **Dashboard → Actions → Add Action → Filter**.
- Select the grouped bar chart as the source.
- Allow selection of a response option to update the stacked chart and table.

Dashboard Title:

"Survey Responses Analysis Dashboard"

Dashboard Components:

Component	Purpose
Grouped Bar Chart	Shows distribution of answers for Question 1
Stacked Bar Chart	Compares response patterns across all questions
Table	Displays individual survey responses

Insight:

The interactive dashboard enables users to analyze response patterns, compare question-wise distributions, and view detailed survey records efficiently. 📊 📋

Set 11

Product Category Analysis

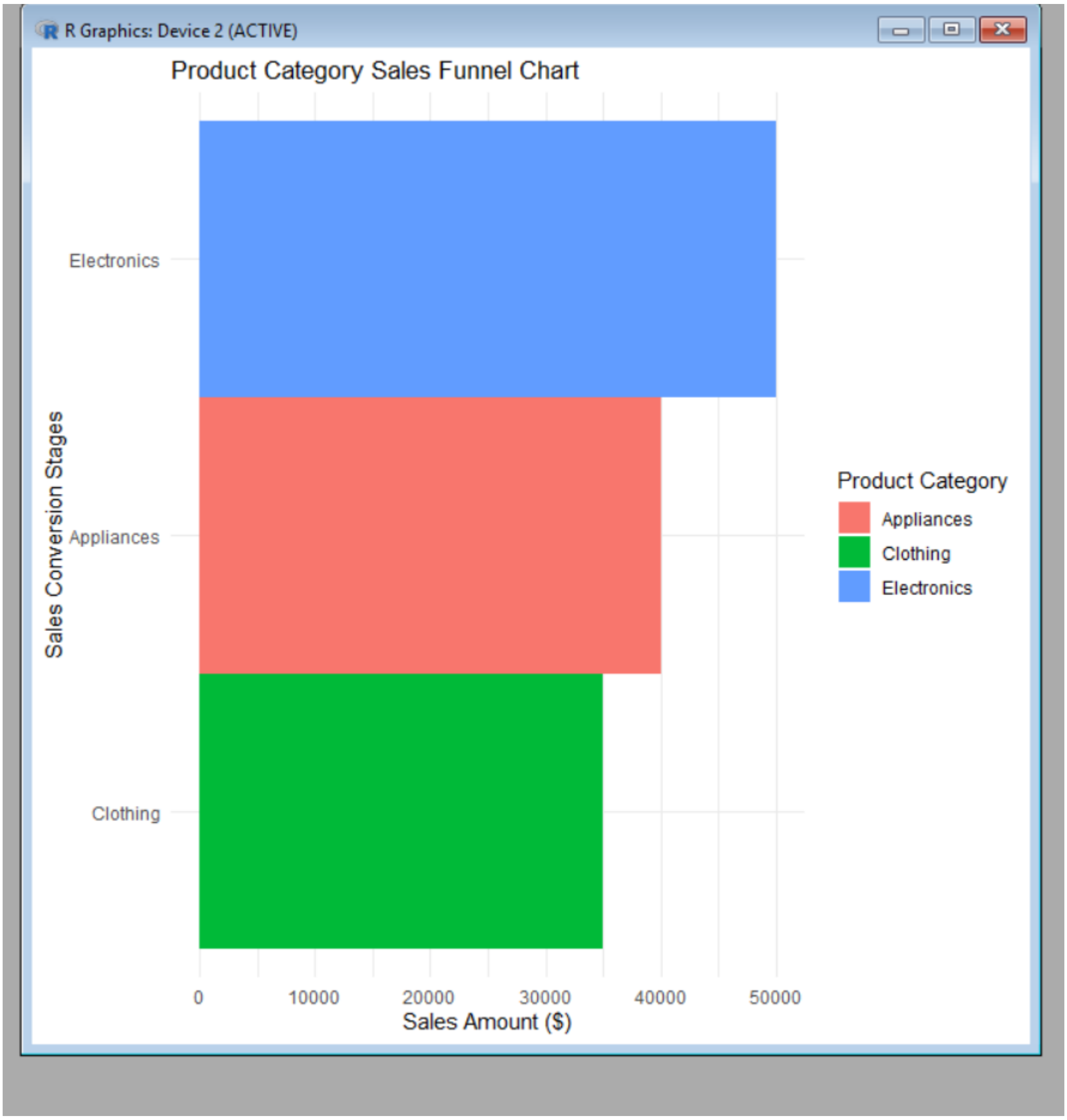
Dataset: Product Categories

1.Generate a funnel chart to analyze the sales conversion process for each product category. Label the

stages and title the chart.

```
# Create dataset

sales <- data.frame(

  Category = c("Electronics", "Clothing", "Appliances"),

  Sales = c(50000, 35000, 40000)

)


# Sort data in descending order

sales <- sales[order(-sales$Sales), ]


# Install and load package
```

```r
library(ggplot2)

# Create funnel chart
ggplot(sales, aes(x = reorder(Category, Sales), y = Sales, fill = Category)) +
  geom_bar(stat = "identity", width = 1) +
  coord_flip() +
  labs(
    title = "Product Category Sales Funnel Chart",
    x = "Sales Conversion Stages",
    y = "Sales Amount ($)",
    fill = "Product Category"
  ) +
  theme_minimal()
```

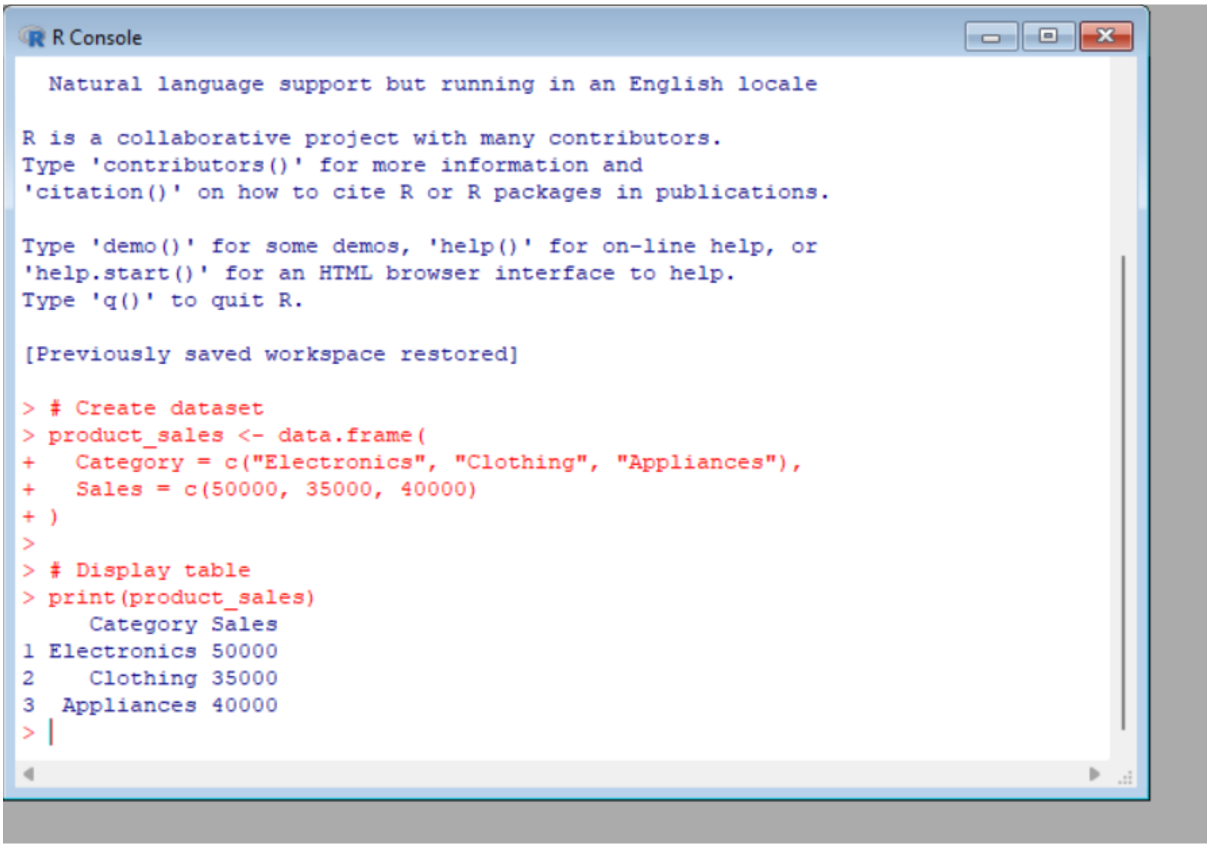


3. **Build a table to display the sales data for each product category. Label the table elements.**

```
# Create dataset
product_sales <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
)
```

Display table

print(product_sales)



3. Develop a Tableau dashboard combining the pie chart, funnel chart, and the table for interactive

exploration of product category data.

Dataset: Product Categories

Category	**Sales (in $)**
Electronics	50000
Clothing	35000
Appliances	40000

Steps to Create Dashboard in Tableau:

1. Create Worksheets

Worksheet 1: Pie Chart (Sales Distribution)

Steps:

1. Import the **Product Categories** dataset into Tableau.
2. Drag **Category** to the **Color** shelf.
3. Drag **Sales** to the **Angle** shelf.
4. Select **Pie Chart** from the visualization options.
5. Drag **Sales** to **Label** to display values.

Chart Title:
"Sales Distribution by Product Category"

Purpose:
Shows the percentage contribution of each product category to total sales.

Worksheet 2: Funnel Chart (Sales Conversion Analysis)

Steps:

1. Drag **Category** to Rows.
2. Drag **Sales** to Columns.
3. Sort sales values in descending order.
4. Select **Bar Chart** and convert it into a funnel-style visualization.
5. Add **Category** and **Sales** values as labels.

Chart Title:
"Product Category Sales Funnel"

Stages:

- Electronics – $50,000
- Appliances – $40,000
- Clothing – $35,000

Purpose:
Displays sales ranking and performance comparison among categories.

Worksheet 3: Sales Data Table

Steps:

1. Drag **Category** to Rows.
2. Drag **Sales** to Text.
3. Select **Text Table**.

Chart Title:
"Product Category Sales Details"

Creating the Tableau Dashboard

1. Click **New Dashboard**.
2. Set dashboard size to **Automatic**.
3. Drag the following sheets into the dashboard:
   - Pie Chart
   - Funnel Chart
   - Sales Data Table

Suggested Layout:

```
-----------------------------------------------
|        Product Category Analysis Dashboard    |
-----------------------------------------------
| Pie Chart            | Funnel Chart       |
| Sales Distribution    | Sales Ranking      |
-----------------------------------------------
|         Product Sales Data Table            |
-----------------------------------------------
```

Add Interactivity:

Filters:

- Add **Category** as a filter.

- Users can select a category to update all charts.

Dashboard Actions:

1. Go to **Dashboard → Actions → Add Action → Filter**.
2. Select Pie Chart as the source.
3. Apply the selection to Funnel Chart and Table.

Dashboard Title:

"Product Category Sales Analysis Dashboard"

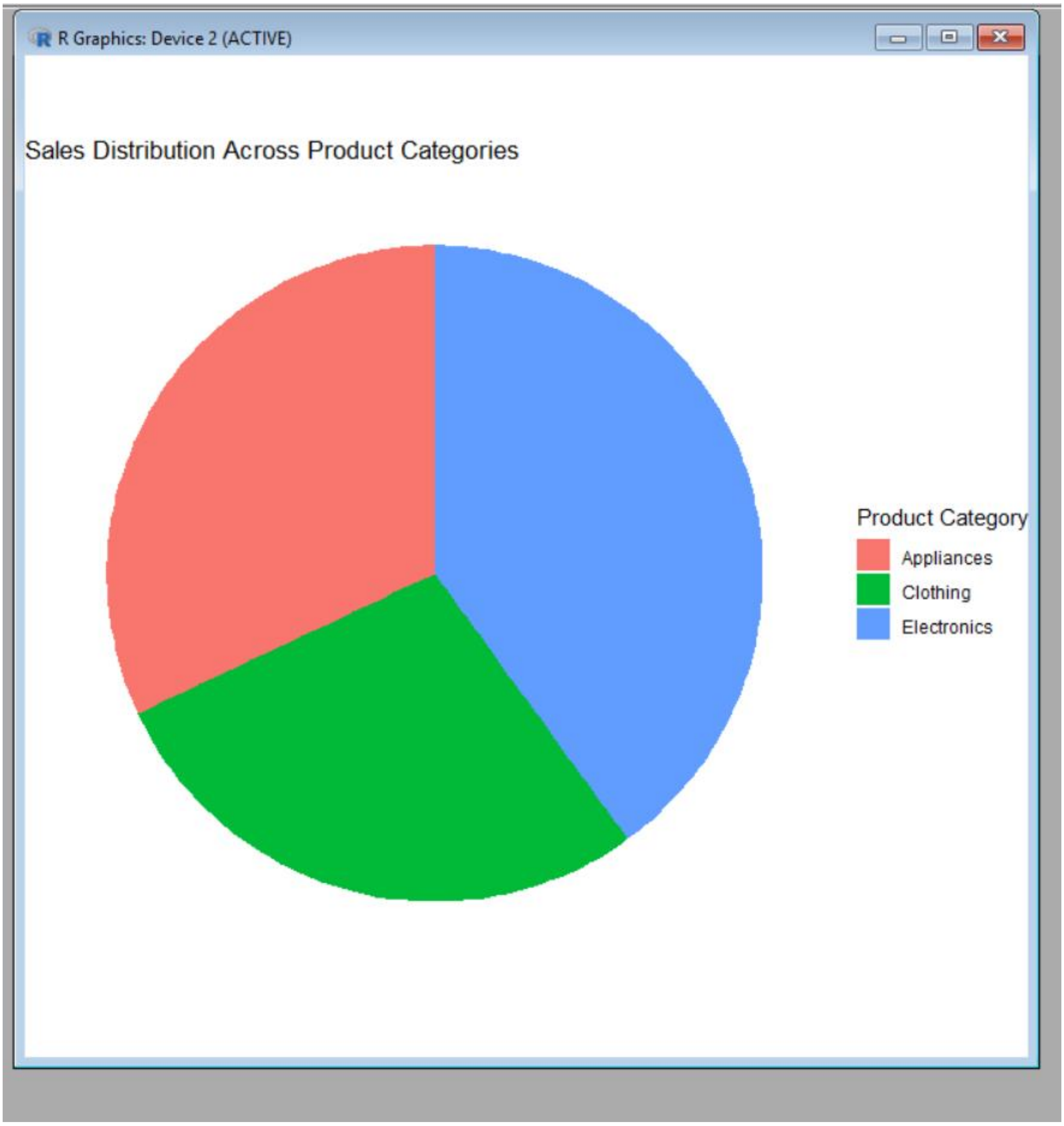
Insights:

- Electronics contributes the highest sales share ($50,000).
- Appliances shows moderate sales performance ($40,000).
- Clothing has the lowest sales contribution ($35,000).
- The interactive dashboard allows users to compare sales visually and explore category-wise performance. 📊 📈

4. **Create a pie chart to represent the distribution of sales across product categories. Include labels.**

```
# Create dataset
product_sales <- data.frame(
 Category = c("Electronics", "Clothing", "Appliances"),
 Sales = c(50000, 35000, 40000)
)

# Load library
library(ggplot2)

# Create pie chart
ggplot(product_sales, aes(x = "", y = Sales, fill = Category)) +
 geom_bar(stat = "identity", width = 1) +
```

```
coord_polar("y") +
labs(
  title = "Sales Distribution Across Product Categories",
  fill = "Product Category"
) +
theme_void()
```



Set 12 :Dataset: Website Traffic

1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the

chart.